Razvan Voicu, PhD

Veteran Technologist & Systems Architect — Scalable Distributed Systems, AI, Payments.
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SUMMARY

Hands-on systems architect and CTO, **30+ years** across fintech, payments, AI-assisted engineering, and high-throughput data engineering. Specializes in re-architecture, performance engineering, cloud cost reduction (**80–90%**), and AI-augmented productivity.

Across TASConnect, GoTo Financial, Meta, Teralytics, and major banks, has repeatedly delivered:

- Cost reduction through multi-tenancy and cloud migration,
- **Kafka/Akka** event-driven systems at scale,
- AI-augmented engineering workflows, and
- Hands-on leadership grounded in first principles engineering.

A **multi-paradigm polyglot engineer** — fluent in Java, Python, Scala, TypeScript, and SQL; has written production code in several others.

Resolutely hands-on, high-impact, architecture-first.

Open to CTO, Principal, Distinguished, and Staff+ roles requiring deep technical leadership and hands-on execution, and to consulting, fractional, and multi-client advisory engagements in architecture, performance, and AI integration.

SIGNATURE ACHIEVEMENTS

- Architected a product line as configurable **finite-state machines**.
    - Replaced per-product hard-coding with **one reusable engine**.
- Scaled a major fintech platform to **~400M monthly transactions**
    - **Rp 55T (~USD 3B) GTV**, **1M+ merchants**.
- Raised uptime to **99.95%**; held **P99 latency at ~500ms**.
- Reduced cloud costs **~80%** at GoTo over 3 years and **~90%** at TASConnect.
- Raised test coverage **15% → 80%**; cut defects **100+ → <10** per release candidate.
- Reduced engineering headcount **160 → 80** and manpower cost **~35%** while improving productivity.
- Led a team operating Meta's critical-path pipelines.
    - **~300TB/day**, **23PB**, **1,000+ node cluster**.
    - On-time delivery: **<15 days/month → 100% of days**.
- Built AI-assisted engineering practice across two organizations
- Authored AI adoption strategy reviewed by Board and independent assessors.
- Converted research-grade probabilistic models into streaming systems; **~3× real-time throughput**.

E X P E R I E N C E

Chief Technology Officer
July 2025 – July 2026

TASConnect · Singapore (registered as *SCV Research & Development Pte Ltd)*

- Re-architected B2B working-capital products (invoice/PO financing on an anchor's credit), writing core parts personally.
- Modeled products as **finite-state machines**, transition logic as data — one backend, all products.
- Multi-tenancy, user-based sharding, stateless serverless tier (PostgreSQL + Lambda):
    - **90% cloud cost reduction**, uptime **<90% → 99%**, **~500ms P99**.
- Delivered mobile single-handedly: frontend to PWA with Android/iOS enclosures.
- Authored an **11-initiative** technology program and AI adoption strategy; reviewed positively by Board and independent reviewers.
    - Designed a hybrid LLM/Mixture-of-Experts and neuro-symbolic AI strategy.
- Hands-on throughout implementation, architecture review, debugging, and live coding.
    - Reduced team size: **40 → 20**.

Director, Head of Engineering, Merchant Services
2019 – 2025

GoTo Financial (Gojek) · Singapore

- Owned payments architecture:
    - **~$3B monthly GTV**, **1M+ merchants**, **~400M monthly transactions.**
    - 1M merchants two months ahead of target.
- Fixed a cross-team API's missing transactions with a dynamic **Postgres advisory lock** and Kafka subscription; adopted team-wide.
- Introduced **Flutter** by building the initial Android/iOS implementation.
    - This work became the foundation of **GoPay Merchant**.
    - Has since grown **53%** and become **market leader**.
- Introduced the **"Add, don't change"** backward compatibility pattern.
- Drove adoption of an existing **infrastructure-as-software** model during an AWS→GCP migration.
- Revamped QA with Cucumber/Gherkin.
    - coverage **15% → 80%**, defects **100+ → <10** per release candidate.
- Led multi-year cloud migrations and infrastructure optimization:
    - **~80%** cost reduction over three years.
- Raised uptime **99% → 99.95%** with automated detection and restart of failing microservices.
    - **P95 <200ms**, **P99 <500ms**.

Adjunct Professor *(part-time)*
2017 – 2023

School of Computing, National University of Singapore

- Designed and taught **Functional Reactive Programming** (**Scala, Kafka**); enrolment **20 → 120+** within three years.
- Taught programming language design, advanced algorithms, data science (R), Python, and software project management.

Data Engineering Manager *2016 – 2019*

Meta · Singapore

- Led **11** engineers across two subteams on ads and monetization analytics.
    - focused on the **China ads market**.
- Operated critical-path pipelines:
    - **~300TB/day**, **23PB** window, **~11,000** pipelines, **1,000+ nodes**, **32-hour** critical path.
- Contributed directly in **Hive SQL, Presto SQL, Scala/Spark.**
- Debugged the scheduler and extended internal tooling.
- Contributed to **critical-path optimization**
    - migration of suitable Hive workloads to Presto/Spark
    - reduced processing **from hours to minutes**.
- As part of a broader engineering effort, on-time delivery: **<15 days/month → 100% of days**.
- Scoped at ~25–30% hands-on; spent **40–50%** coding and debugging.

Data Engineering Lead *2015 – 2016*

Teralytics · Singapore

- Led four engineers turning PhD-built **Bayesian HMM** models into production streaming inference for Singapore's **LTA**.
- **Near real-time output** (10 min delay).
- Converted multi-pass batch algorithms into online, event-driven functional transformations.
    - **Scala, Kafka, Akka, Mesos**.
- Optimized for streaming: **~3× real-time throughput** — a day of input in **under eight hours**.

Earlier Career — Financial Systems (VP Level) *Pre-2015*

Bank of America · J.P. Morgan · Standard Chartered Bank

- **Standard Chartered** (VP, 2008–2011) — **30+ Haskell** trading apps: FX derivatives, baskets, indices, structured products; Monte Carlo on an **18-machine** cluster.
- **J.P. Morgan** (VP, 2011–2013) — Led five of a 40-person org on custody asset-servicing
    - **Microservices/AMQP**, Java, C#, WPF.
- **Bank of America Merrill Lynch** (VP, 2013–2015) — Led seven on a trade allocation engine.
    - Matching modeled as **NP-complete Subset Sum** with pricing-tolerance heuristics
    - **Zero production incidents**.

EDUCATION

PhD, Computer Science *1998 – 2004*

School of Computing, National University of Singapore

- **Valedictorian, class of 2004** · **Best thesis award** · IDA gold medal & cash award · Dean's graduate award (2000) · GPA (NUS CAP) 5.0 / 5.
- **Bell Labs, New Jersey:** multiple internships.
  - Implemented the register allocator for an ARM compiler backend in Standard ML (SML/NJ).
- Lectured across the undergraduate CS curriculum: programming, algorithms, operating systems, software engineering, compilers, distributed systems, formal methods.
- Built a reputation for making **demanding subjects accessible**.
  - Subsequently central to **mentoring engineers** and **communicating architecture**.
- Author of **14** peer-reviewed publications in program verification and automated reasoning.

Bachelor of Computer Engineering *1987 – 1991*

Asachi Technical University · Romania

- Valedictorian, class of 1991 · Average overall mark: 9.96 / 10.

CORE COMPETENCIES

- **Payments & Merchant Platforms** — $3B+ GTV, 400M+ transactions/month; regulatory-aware design.
- **Software Architecture** — reactive, event-driven, multi-tenant, FSM-based; AWS/GCP/AliCloud, Kubernetes, serverless.
- **Artificial Intelligence** — AI-assisted engineering, Symbolic AI.
- **Distributed Systems** — high-throughput reactive systems; Kafka/Akka; large-scale data platforms.
- **Engineering Leadership** — teams to 160; organizational redesign, quality engineering, cost optimization.
- Java, Kotlin, Scala/Spark, Golang, Python, TypeScript, Swift, Flutter/Dart, C/C++, Linux, Android/iOS.

LANGUAGES

- English — full professional
- French — professional working
- Romanian — native
- German — limited working
- Italian — elementary
- Indonesian — conversational

HOBBIES

Motorbike touring, hobby electronics, IoT gadgets built from scratch.